MATH 3603–Homework 1

Due: 08-Feb-2026

1. Prove the following formula using induction:

$$P(E_1 \cup E_2 \cup \cdots \cup E_n) = \sum_i P(E_i) - \sum_{i<j} P(E_i \cap E_j) + \sum_{i<j<k} P(E_i \cap E_j \cap E_k)$$

$$- \sum_{i<j<k<l} P(E_i \cap E_j \cap E_k \cap E_l) + \cdots + (-1)^{n+1} P(E_1 \cap E_2 \cap \cdots \cap E_n).$$

2. For events E_1, E_2, $\cdots$, E_n, show that

$$P(E_1 \cap E_2 \cap \cdots \cap E_n) = P(E_1)P(E_2|E_1)P(E_3|E_1 \cap E_2) \cdots P(E_n|E_1 \cap \cdots \cap E_{n-1}).$$

3. Prove Boole's inequality:

$$P(\cup_{i=1}^n E_i) \leq \sum_{i=1}^n P(E_i).$$

hint: First consider the simple case when $n = 2$. The idea is to try to partition $\cup_{i=1}^n E_i$ into disjoint (mutually exclusive) set F_i's, where $F_1 = E_1$ and $F_i = E_i \cap (\cap_{j=1}^{i-1} (E_j)^c)$.

4. Only two factories produce rulers. 20 percent of the rulers from factory I and 5 percent from factory II are defective. Factory I produces twice as many rulers as factory II each week.

(i): What is the probability that a ruler, randomly chosen from a week's production, is satisfactory?

(ii): If the chosen ruler is defective, what is the probability that it came from factory I?

5. If events A, B, C are independent, prove the following statements

(i) A is independent of B^c,

(ii) A is independent of $B \cap C$,

(iii) A is independent of $B \cup C$.